obtain the wave equation for dielectric media.

$$\vec{E} = \vec{A} e^{i(\vec{K}\cdot\vec{r}) - \omega t}$$

$$\vec{B} = \mu \vec{H}$$

↳ ley de Faraday

$$\left.\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}\right\} \Rightarrow \nabla \times \left(\vec{A} e^{i(\vec{K}\cdot\vec{r} - \omega t)}\right) = -\frac{\partial}{\partial t}(\vec{B})$$

Integrar respecto al tiempo.

$$\nabla \times \left(\vec{A}\, e^{i(\vec{K}\cdot r - \omega t)}\right) = -\frac{\partial}{\partial t}\mu \vec{H}$$

$$\nabla \times (\phi \vec{A}) = \phi\, \nabla \times \vec{A} +$$

$$= -\mu \frac{\partial}{\partial t}\vec{H}$$

$$\left.\frac{d}{dt}(AB) = A\frac{d}{dt}B + B\frac{d}{dt}A\right\}$$

Haciendo los calculos:

Tenemos...

$$i(\vec{K}\times\vec{A})\, e^{i(\vec{K}\cdot\vec{r} - \omega t)} = -\mu \frac{\partial}{\partial t}\vec{H}$$

Por teo. Fundamental del cálculo.

$$i(\vec{K}\times\vec{A}) \int e^{i(\vec{K}\cdot\vec{r} - \omega t)}\, dt = -\mu \int \frac{\partial}{\partial t}\vec{H}\, dt$$

$$i(\vec{k}\times\vec{A})\int e^{i(\vec{k}\cdot\vec{r}-wt)}\,dt = -\mu\,\vec{H}$$

$$\Rightarrow \quad \vec{H} = -\frac{i}{\mu}(\vec{k}\times\vec{A})\int \underbrace{e^{i(\vec{k}\cdot\vec{r}-wt)}}\,dt$$

$$e^{i(\vec{k}\cdot\vec{r}-wt)} = e^{i\vec{k}\cdot\vec{r}}\,e^{-iwt}$$

$$\Rightarrow \quad \vec{H} = -\frac{i}{\mu}(\vec{K}\times A)\left[e^{i\vec{K}\cdot\vec{r}}\int e^{-iwt}\,dt\right]$$

$u = -iwt$

$du = -iw\,dt \quad \Rightarrow \quad dt = -\frac{du}{iw}$

$$= -\frac{i}{\mu}(\vec{K}\times\vec{A})\left[e^{i\vec{K}\cdot\vec{r}}\left(-\frac{1}{iw}e^{-iwt}\right)\right] + C$$

$$\vec{H} = -\frac{i}{\mu}(\vec{K}\times\vec{A})\left[-\frac{1}{iw}\,e^{i(\vec{K}\cdot\vec{r}-wt)}\right] + C$$

$$\vec{H} = \frac{1}{\mu w}\,\vec{K}\times\vec{A}\,e^{i(\vec{K}\cdot\vec{r}-wt)} + C$$

2. Ley de Gauss